



The University of Faisalabad

Course Code	Course Title	Credit Hours	Contact Hrs/Week	Total Lectures Allocated
CS-225	Operating System	3(2-1)	5	48

Discipline	Semester no	Commencement of Semester	Mid Semester Examination	Semester Termination Date
BSCS+BSSE+BSAI	IV			

COURSE LEVEL

Undergraduate

PREREQUISITES

Data Structures and Algorithms

COURSE DESCRIPTION:

This course will introduce the core concepts of operating systems, such as processes and threads, scheduling, synchronization, memory management, file systems, input and output device management and security. The course will consist of assigned reading, weekly lectures, a midterm and final exam, and a sequence of programming assignments. The goal of the readings and lectures is to introduce the core concepts. The goal of the programming assignments is to give students some exposure to operating system code. Students are expected to read the assigned materials prior to each class, and to participate in in-class discussions.

COURSE LEARNING OBJECTIVES

At the end of the course, you should be able to:

- Grasp the working of an operating system
- Analyze the synchronization process
- Identify the working methodology of multi-threaded applications
- Determine the reasons of deadlocks in an operating system
- Find the management of different type of memories in the computer system
- Compare different file systems being used in different operating systems

KEY SKILLS AND ATTRIBUTES TO BE DEVELOPED BY THE COURSE

Knowledge, understanding and application skills

TECHNOLOGY REQUIREMENTS

-Computer System
-Learning Management System
-Supporting Software (e.g. MS Office, Acrobat Reader, Web Browser, Media player)
- Internet Facility
- Digital Library Access
- Visual Studio

REQUIRED LEARNING RESOURCES

Text Book(s):	- Abraham Silberschats, Peter Baer Galvin, Gerg Gagne, Operating System Concepts, 9 th Edition, Wiley
Reference Book(s):	- Applied Operating Systems Concepts, 7th Edition, Silberschatz A., Peterson, J.L., & Galvin P.C. 2004.

WEEKLY PLAN			
Weeks	Lecture No.	Title of Chapter	Contents
Week 1	1	Introduction to OS	Computer System Organization, Computer System Architecture, Operating-System Structure
	2		Evolution of multi-user systems, Single User and Multiuser operating system
Week 2	3	Introduction to OS	Types of Operating Systems: Batch Processing, Time Sharing Systems, Batch system
	4		Operating-System Operations: Process Management, Memory Management,Storage Management, Protection and Security
Week 3	5	Operating - System Structures	Operating-System Services,User and Operating-System Interface
	6		System Calls, Types of System Calls
Week 4	7	Operating - System Structures	Operating-System Design and Implementation
	8		Operating-System Debugging, System Boot
Week 5	9	Process Management	Process Concept: Process states, threads.
	10		Process Scheduling: Scheduling Queues, Schedulers, Context Switch Operations on Processes
Week 6	11	Process Synchronization	The Critical-Section Problem, Peterson’s Solution,
	12		Synchronization Hardware, Mutex Locks,Semaphores: Semaphore Usage, Deadlocks and Starvation
Week 7	13	Process Synchronization	Classic Problems of Synchronization: The Bounded-Buffer Problem, The Readers–Writers Problem , The Dining-Philosophers Problem
	14	CPU Scheduling	Basic Concepts: CPU–I/O Burst Cycle, CPU Scheduler, Preemptive Scheduling non preemptive scheduling.
Week 8	15	CPU Scheduling	Scheduling Criteria, Scheduling Algorithms: First-Come, First-Served Scheduling, Shortest-Job-First Scheduling
	16		Priority Scheduling, Round-Robin Scheduling, Multilevel Queue Scheduling, Multilevel Feedback Queue Scheduling
Week 9	MID EXAMS		
Week 10	17	Deadlocks	System Model, Deadlock Characterization, Necessary Conditions, Resource-Allocation Graph
	18		Methods for Handling Deadlocks, Deadlock Prevention, Mutual Exclusion, Hold and Wait Methods for Handling Deadlocks,
Week 11	19	Deadlocks	Deadlock Avoidance, Safe State, Resource-Allocation-Graph Algorithm, Banker’s Algorithm, Resource-Request Algorithm
	20		Deadlock Detection, Recovery from Deadlock, Resource Preemption,
Week 12	21	Memory Management	Basic Hardware, Address Binding, Logical Versus Physical Address Space,Dynamic Loading, Dynamic Linking and Shared Libraries,
	22		Swapping, Contiguous Memory Allocation: Memory Protection, Memory Allocation, Fragmentation,Segmentation, Paging,Basic Method, Hardware Support, Protection,
Week 13	23	Virtual Memory	Demand Paging, Basic Concepts, Performance of Demand Paging,Copy-on-Write, Page Replacement, Basic Page Replacement
	24		FIFO Page Replacement, Optimal Page Replacement, LRU Page Replacement
Week 14	25	Mass-Storage Structure	Overview of Mass-Storage Structure, Disk Structure, Disk Attachment, Disk Formatting,Boot Block
	26		Disk Scheduling: FCFS Scheduling, SSTF Scheduling, SCAN Scheduling, C-SCAN Scheduling, LOOK Scheduling
Week 15	27	File -System Interface	File Concept, File Attributes, File Operations, File Types
	28		File Structure, Internal File Structure, Access Methods: Sequential Access, Direct Access

Week 16	29	File -System Interface	Directory and Disk Structure, Storage Structure, Directory Overview, Single-Level Directory, Two-Level Directory, File Sharing
Week 17	31	Protection and Security	Goals of Protection, Principles of Protection
	32		Domain of Protection, Access Matrix,Access Control, Revocation of Access Rights
Week 18	FINAL EXAMS		

COURSE ASSESSMENT

-Homework exercise in the form of supplementary reading materials will be given to students according to the course progress; homework will be not marked and will not be counted towards the course assessment.
-2 quiz and (2-3) assignments will be given during the semester
-1 mid-term exam and 1 final exam will be performed.

ASSESSMENT SCHEME

Internal Assessment:	35%
Mid Semester Examination:	25%
Final Semester Examination:	40%

GUIDELINES FOR SUBMISSION OF ASSIGNMENTS

Read the assignment requirements carefully, including any specific instructions on length, formatting, style, etc., provided for every assignment. If no specific instructions are given, here are some general suggestions for you to follow:
-Include sufficient identification on every assignment submitted (e.g., your name; the course name, course code, and the assignment number or title).
-For assignments with multiple questions, identify the question number and restate each assignment question before providing your answer.
-Review your assignment before submitting it to make sure you have completed the assignment in full. Also, carefully proofread your work for spelling and grammatical errors that could affect your grade.
-Submit assignment on LMS till due date.
-Also, unless you have registered for an extension, assignments submitted after the due date, will be returned to you ungraded.

MARKS DISTRIBUTION AND GRADING MODEL

Grade	Grade Points	Equivalent Percentage
A	4	85 - 100
A-	3.7	80 - 84
B+	3.3	75 - 79
B	3	70 - 74
B-	2.7	65 - 69
C+	2.3	60 - 64
C	2	58 - 59
C-	1.7	55 - 57
D	1	50 - 54
F	0	Below 49

ACADEMIC HONESTY

Students are expected to conduct themselves in accordance with the highest ethical standards of the Profession of Engineering and Technology and evince academic integrity in all their pursuits and activities at the university. As such, in accordance with the general academic regulations and requirements of the University of Faisalabad, students are reminded that plagiarism or any other form of cheating in examinations, assignments, laboratory reports or term tests is subject to serious academic penalty (e.g. cancellation of exam/assignment/quiz). A student found guilty of contributing to cheating in examinations or term assignments is also subject to serious academic penalty.

DISCLAIMER

This course outline is tentative; I reserve the right to change the deadlines, readings, or assignments during the semester.

Note:-The above course outline covers all contents of the subject as prescribed by HEC and respective accreditation Council.

Instructor's name: Ms Tahmina Asghar

Signature & Date_____

HOS/COS_____